



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**BACHELOR OF SCIENCE COMPUTER (DATA
ENGINEERING)**

SECP3723 - SYSTEM DEVELOPMENT TECHNOLOGY

SEMESTER 2024/2025 – 1

SECTION 01

COURSE REGISTRATION SYSTEM

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1.0 SYSTEM INTRODUCTION

OVERVIEW

The Course Registration System aims to simplify and enhance the course enrollment process for students, while also offering key features for lecturers and administrators. This system modernizes previous methods with a more intuitive interface and includes security upgrades along with strong backend management. Built with PHP, MySQL, and Bootstrap, it guarantees accessibility, reliability, and user-friendliness for everyone involved.

SYSTEM GOALS

The primary goal of this system is to create a new, user-friendly platform for course registration that enhances usability while maintaining essential administrative controls. It provides lecturers with access to their assigned course details, allows students to register, modify, and cancel their course enrollments, and enables administrators to manage enrollment processes and course information effectively.

KEY FEATURES

1. Student Key Features/Functions
 - a) Register courses
 - b) View current and previous semester courses
 - c) Search course
 - d) Modify registration
 - e) Cancel registration
 - f) Edit profile
2. Lecturer Key Features/Functions
 - a) View assigned courses for each semester
 - b) View student list for each course
 - c) View course details
 - d) View student details
3. Admin Key Features/Functions
 - a) Add new course details with number of maximum students per course
 - b) Modify course details
 - c) Delete course
 - d) Amend or cancel registration
 - e) Dashboard

SECURITY AND RELIABILITY ENHANCEMENTS

The following measures are implemented to ensure system security and maintain data integrity:

- Password Protection: User passwords are either hashed or encrypted to enhance security.
- SQL Injection Prevention: Secure coding practices are employed to prevent unauthorized access to the database.
- User Authentication & Verification: Login credentials are verified to prevent unauthorized access.

TECHNOLOGY AND DEVELOPMENT APPROACH

The system is built using the following technologies:

- PHP: for backend programming and logic.
- Database: MySQL (to manage registration, course, and user data).
- Frontend: Bootstrap (to create a responsive user interface).
- Development Methodology: Agile, ensuring iterative development and continuous improvements.

DEPLOYMENT AND HOSTING

To ensure accessibility, the Course Registration System will be hosted on a dedicated server. The final application package will include all system files, a MySQL database script, and detailed setup instructions for live deployment.

2.0 SYSTEM DESIGN

USE CASE

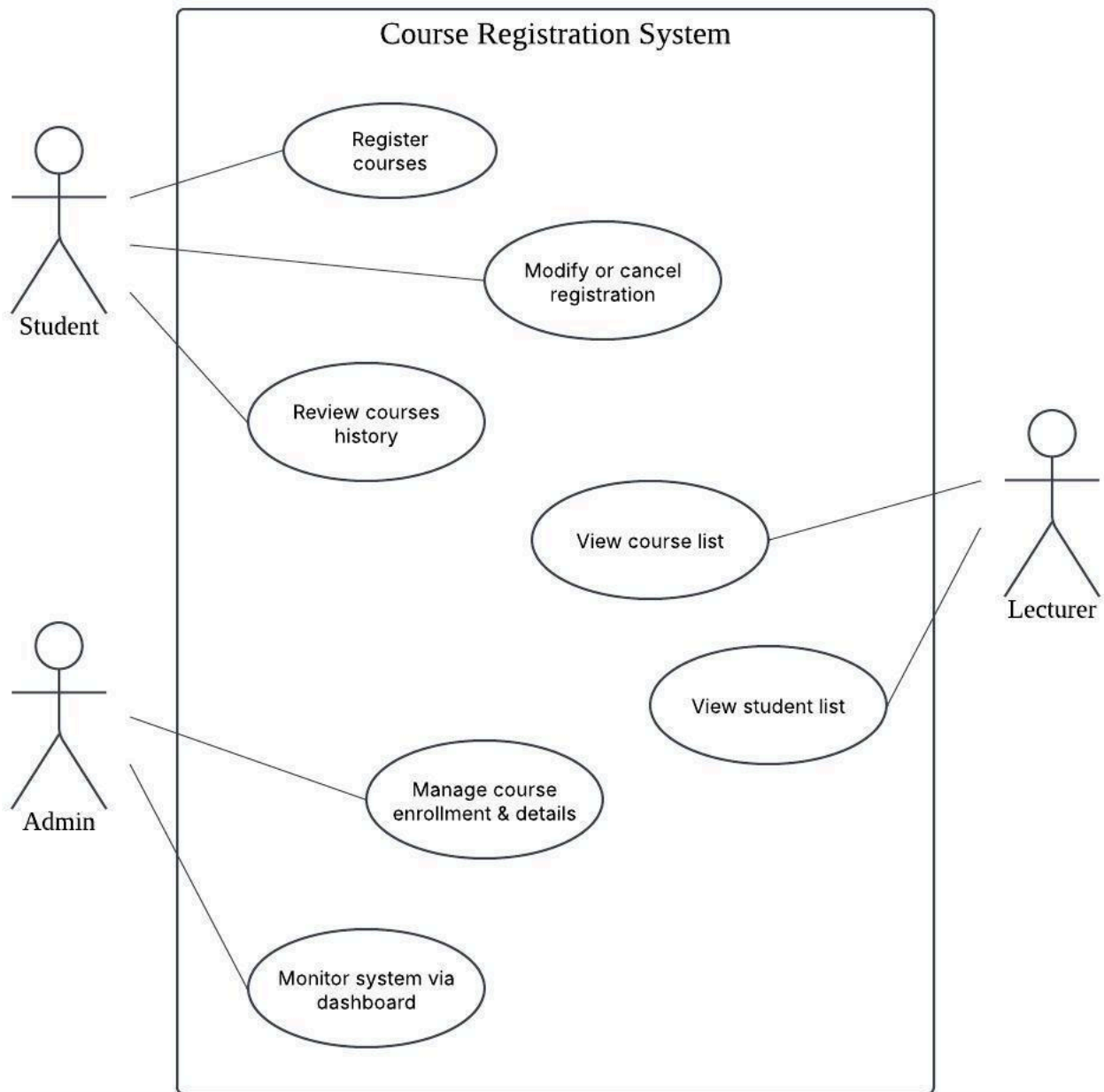


Figure 2.1.1 Use Case Diagram for Course Registration System

ENTITY RELATIONSHIP DIAGRAM (ERD)

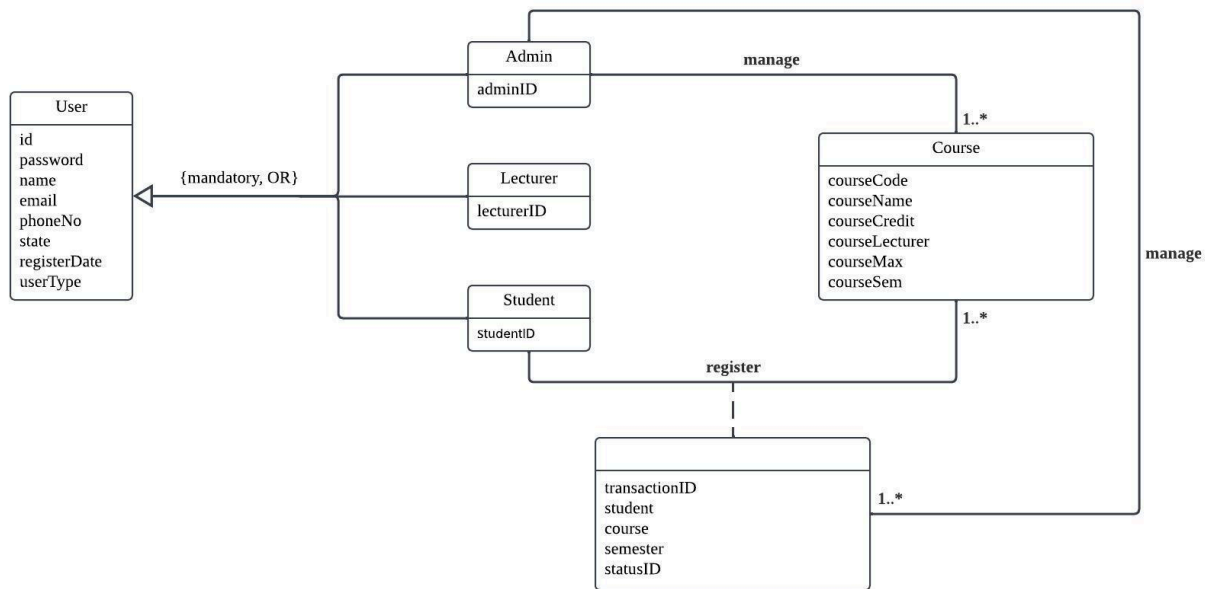


Figure 2.2.1 Conceptual Entity Relationship Diagram for Course Registration System

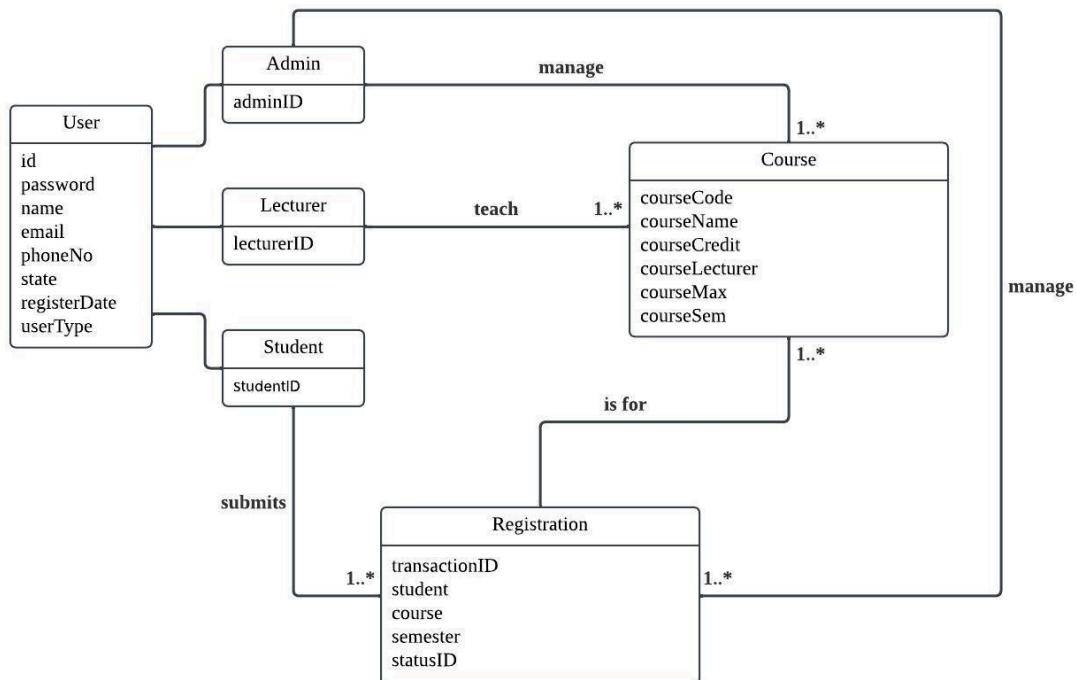


Figure 2.2.2 Logical Entity Relationship Diagram for Course Registration System

RELATIONAL SCHEMA

1. tb_course (c_code, c_name, c_credit, c_lec, c_sem, c_max)

Primary Key : c_code

Foreign Key : c_lec REFERENCES tb_user(u_sno)

This table already achieves 3NF normalization since there's no transitive dependencies. Both c_sem and c_max depend totally on the primary key itself, rather than course scheduling.

2. tb_registration (r_tid, r_student, r_course, r_sem, r_status)

Primary Key : r_tid

Foreign Key : r_student REFERENCES tb_user(u_sno)

Foreign Key : r_course REFERENCES tb_course(c_code)

Foreign Key : r_status REFERENCES tb_status(s_id)

This table already achieves 3NF normalization since there's no transitive dependencies. Both c_sem and r_sem have different purposes in different tables, therefore they won't clash.

3. tb_status (s_id, s_desc)

Primary Key : s_id

This table already achieves 3NF normalization since there's no transitive dependencies.

4. tb_user (u_sno, u_pwd, u_email, u_name, u_contact, u_state, u_reg, u_utype)

Primary Key : u_sno

Foreign Key : u_utype REFERENCES u_utype(t_id)

This table already achieves 3NF normalization since there's no transitive dependencies.

5. u_utype (t_id, t_desc)

Primary Key : t_id

This table already achieves 3NF normalization since there's no transitive dependencies.

3.0 DEVELOPMENT SOFTWARE, LANGUAGE, TECHNOLOGY AND TOOLS

DEVELOPMENT SOFTWARE

The project utilizes **XAMPP**, a well-known local server solution that includes PHP, Perl, MySQL, and Apache. Developers can test and build the system locally using XAMPP before deployment. It provides a controlled environment for running web applications, server-side scripting, and managing databases.

PROGRAMMING LANGUAGE

PHP serves as the primary server-side scripting language for the system's development, allowing for dynamic content creation and interaction with databases. JavaScript complements PHP by enhancing front-end interactions, thereby improving the overall user experience. To facilitate efficient data management within MySQL, SQL is utilized for executing database queries and retrieving data.

TECHNOLOGY AND FRAMEWORKS

1. **Bootstrap & Bootstrap:** Bootstrap provides pre-styled themes to enhance the visual of the system without CCs extensive custom, while Bootstrap ensures a responsive and mobile-friendly layout. Together, they create a modern and user-friendly interface.
2. **MySQL (PHPMyAdmin):** The database management system that efficiently stores and manages data related to registrations, courses, and users.

DEVELOPMENT TOOLS

1. **Sublime Text:** This lightweight but powerful code editor is ideal for writing and managing PHP, JavaScript, and SQL code for the system. Its features, such as multi-line editing and syntax highlighting, enhance coding productivity.
2. **LucidChart:** A diagramming tool that helps create Use Case Diagrams, ERDs (Entity-Relationship Diagrams), and other visual representations of system designs. This tool is beneficial for understanding system relationships and planning database structures.

4.0 DATABASE DESIGN

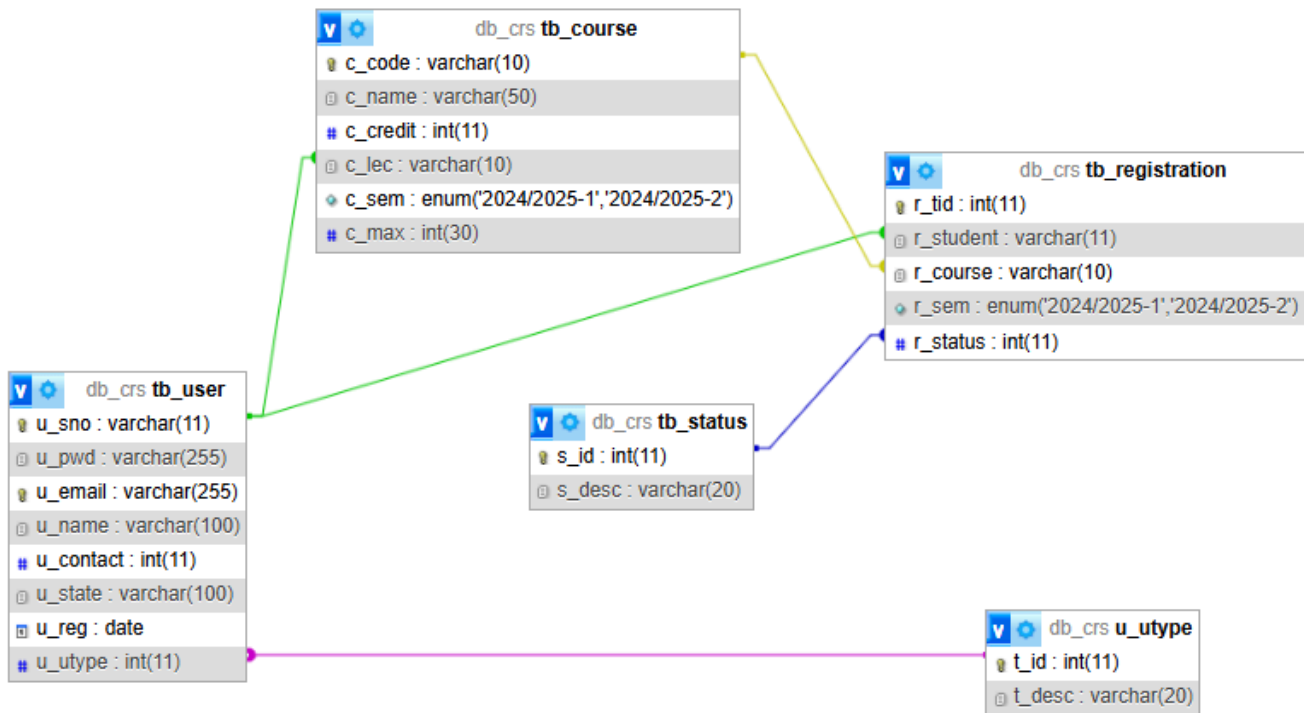


Figure 4.0.1 Designer Diagram for Course Registration System

Table	Action	Rows	Type	Collation	Size
tb_course	★ Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_general_ci	32.0 KiB
tb_registration	★ Browse Structure Search Insert Empty Drop	9	InnoDB	utf8mb4_general_ci	64.0 KiB
tb_status	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB
tb_user	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	48.0 KiB
u_utype	★ Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB
5 tables	Sum	33	InnoDB	utf8mb4_general_ci	176.0 KiB

Figure 4.0.2 Table Diagram for Course Registration System
(rows size may vary due to add/delete data)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	c_code	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
2	c_name	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
3	c_credit	int(11)			No	None			Change Drop More
4	c Lec	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
5	c Sem	enum('2024/2025-1', '2024/2025-2')	utf8mb4_general_ci		No	None			Change Drop More
6	c_max	int(30)			No	0			Change Drop More

Figure 4.0.3 Table Diagram for tb_course

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	r_tid 🔑	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	r_student 🔑	varchar(11)	utf8mb4_general_ci		No	None			Change Drop More
3	r_course 🔑	varchar(10)	utf8mb4_general_ci		No	None			Change Drop More
4	r_sem	enum('2024/2025-1', '2024/2025-2')	utf8mb4_general_ci		No	None			Change Drop More
5	r_status 🔑	int(11)			No	None			Change Drop More

Figure 4.0.4 Table Diagram for tb_registration

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	s_id 🔑	int(11)			No	None			Change Drop More
2	s_desc	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Figure 4.0.5 Table Diagram for tb_status

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	u_sno 🔑	varchar(11)	utf8mb4_general_ci		No	None			Change Drop More
2	u_pwd	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
3	u_email 🔑	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
4	u_name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
5	u_contact	int(11)			No	None			Change Drop More
6	u_state	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
7	u_reg	date			No	None			Change Drop More
8	u_utype 🔑	int(11)			No	None			Change Drop More

Figure 4.0.6 Table Diagram for tb_user

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	t_id 🔑	int(11)			No	None			Change Drop More
2	t_desc	varchar(20)	utf8mb4_general_ci		No	None			Change Drop More

Figure 4.0.7 Table Diagram for u_utype

5.0 DEVELOPMENT STEPS

The Course Registration System was designed through an iterative process that started with a requirement analysis to pinpoint the key features needed by administrators, lecturers, and students. In addition to addressing database requirements and system constraints, security measures such as authorization, authentication, and protection against SQL injection were put in place. During the system design phase, Entity Relationship Diagrams (ERDs) and relational schemas ensured that the database was organized and standardized, while Use Case Diagrams illustrated how users would interact with the system.

MySQL tables were created using phpMyAdmin, ensuring proper restrictions for data consistency during the database and backend development process. Key features, including course registration, modification, and cancellation, along with session management for role-based access and authentication, were implemented using PHP. To protect user data, security measures such as SQL query sanitization and password encryption were employed. Meanwhile, frontend development focused on building a responsive interface with JavaScript and Bootstrap to enhance usability and provide real-time validation.

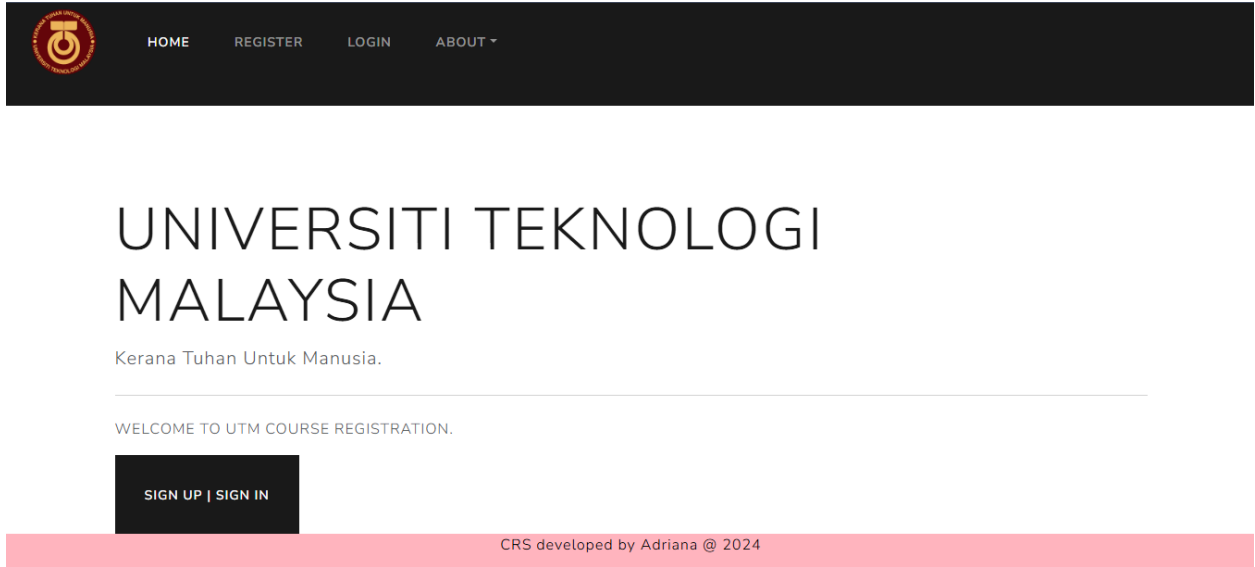
After developing the frontend and backend components, I ensured that the modules worked together smoothly through system integration and testing. The database verified transactions and overall functionality with unit and integration tests. Before finalizing the hosting setups, the system is tested in a local XAMPP environment. In the end, a comprehensive review and documentation process resulted in user manuals, ensured system reliability, and provided detailed documentation for future use.

These systematic development procedures guaranteed a secure, efficient, and user-friendly course registration system that met all functional and technical requirements.

6.0 SYSTEM INTERFACE FOR ALL USERS

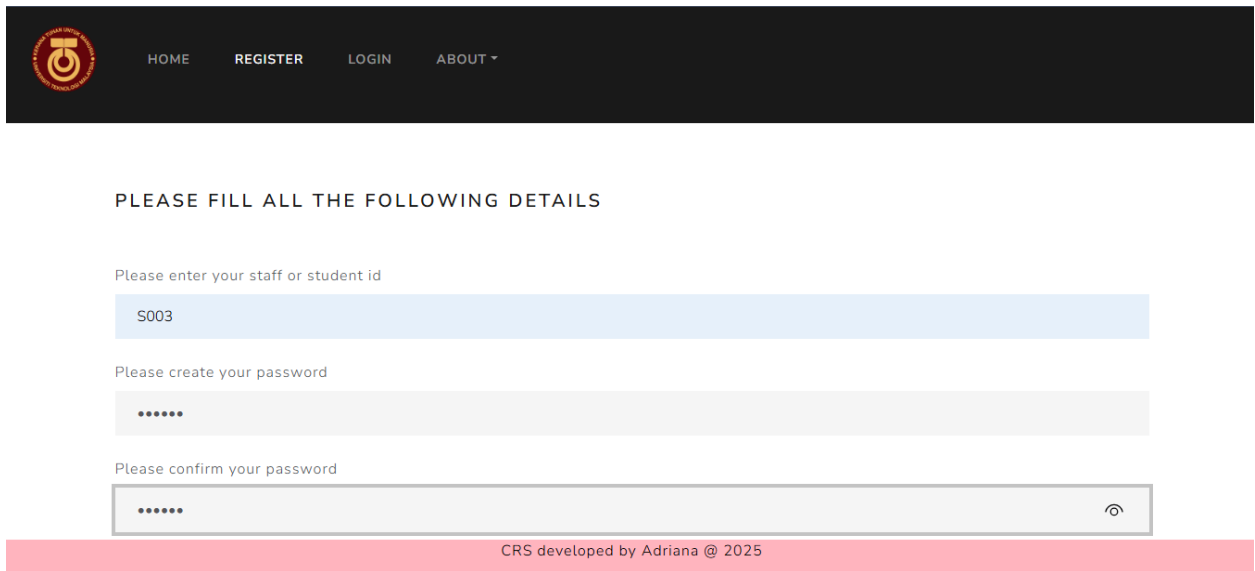
For Student ;

1. At the guest homepage, select the Register section for new users.



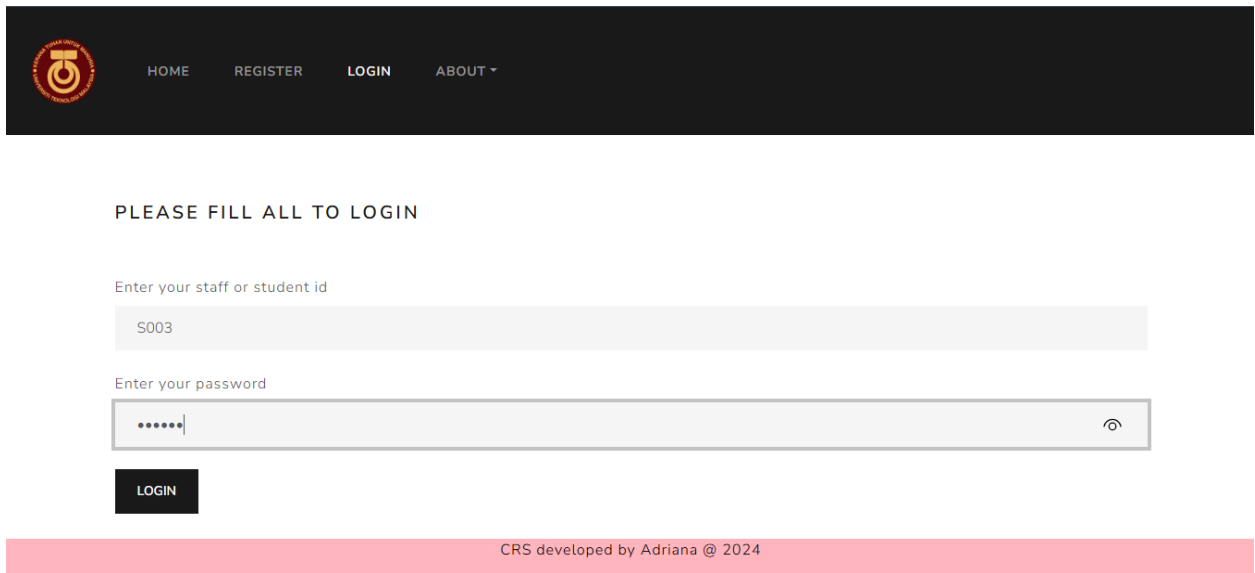
The screenshot shows the homepage of the UTM Course Registration system. At the top is a dark navigation bar with the UTM logo on the left and links for HOME, REGISTER, LOGIN, and ABOUT on the right. Below the navigation bar, the text "UNIVERSITI TEKNOLOGI MALAYSIA" is displayed in a large, bold, sans-serif font. Underneath this, the motto "Kerana Tuhan Untuk Manusia." is written in a smaller font. A horizontal line separates the header from the main content area. Below the line, the text "WELCOME TO UTM COURSE REGISTRATION." is centered. A dark button with the text "SIGN UP | SIGN IN" is positioned to the left of the center. At the bottom of the page, a pink footer bar contains the text "CRS developed by Adriana @ 2024" on the right side.

2. Fill in the Registration Form as new users.



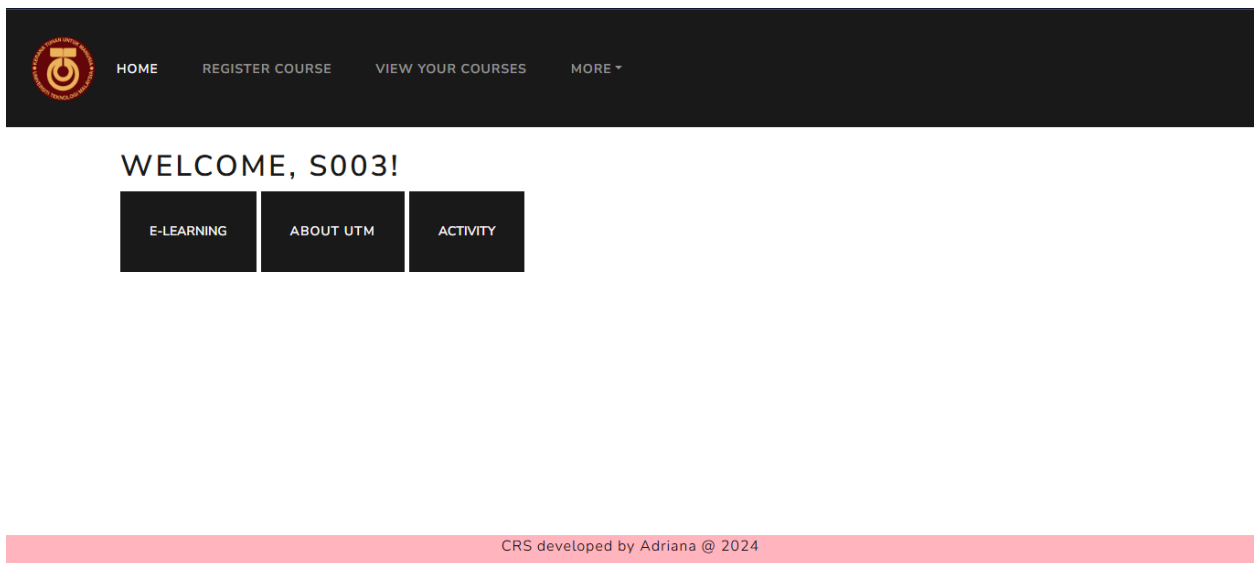
The screenshot shows the registration form page. It features a dark navigation bar at the top with the UTM logo and links for HOME, REGISTER, LOGIN, and ABOUT. The main heading "PLEASE FILL ALL THE FOLLOWING DETAILS" is centered. Below this, there are three input fields: 1. A text field labeled "Please enter your staff or student id" with the value "S003" entered. 2. A password field labeled "Please create your password" with masked characters ".....". 3. A password field labeled "Please confirm your password" with masked characters "....." and a toggle icon on the right. At the bottom, a pink footer bar contains the text "CRS developed by Adriana @ 2025" on the right side.

3. The page will redirect to the Login page once registration is successful.



The screenshot shows a login page with a dark blue header. On the left is the UTM logo. To its right are navigation links: HOME, REGISTER, LOGIN, and ABOUT with a dropdown arrow. The main content area has the heading "PLEASE FILL ALL TO LOGIN". Below this are two input fields: "Enter your staff or student id" with the value "S003" and "Enter your password" with masked dots and a toggle icon. A black "LOGIN" button is below the password field. A pink footer bar at the bottom contains the text "CRS developed by Adriana @ 2024".

4. This is the homepage of every student.



The screenshot shows a student homepage with a dark blue header. On the left is the UTM logo. To its right are navigation links: HOME, REGISTER COURSE, VIEW YOUR COURSES, and MORE with a dropdown arrow. The main content area has the heading "WELCOME, S003!". Below this are three black buttons: "E-LEARNING", "ABOUT UTM", and "ACTIVITY". A pink footer bar at the bottom contains the text "CRS developed by Adriana @ 2024".

5. Select the Register Course section to register the course. Search section is placed at the top for students to find the courses they want.

 [HOME](#) [REGISTER COURSE](#) [VIEW YOUR COURSES](#) [MORE ▾](#)

COURSE REGISTRATION FORM

Search courses [SEARCH](#)

Please select current semester

2024/2025-2 ▾

Please select an available course

SECP3223 - Data Analytics Programming ▾

[REGISTER](#) [CLEAR FORM](#)

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6. Once the Register button is clicked, students will be transferred to the View Your Courses page. Here, students can modify or cancel their registration.

 [HOME](#) [REGISTER COURSE](#) [VIEW YOUR COURSES](#) [MORE ▾](#)

VIEW YOUR COURSES

You can only modify next semester course registration.

Select Semester:

Semester 2024/2025-2 ▾

TRANSACTION ID	SEMESTER	COURSE	COURSE NAME	STATUS	ACTIONS
36	2024/2025-2	SECP3223	Data Analytics Programming	Accepted	MODIFY CANCEL

CRS developed by Adriana @ 2024

7. At the More section dropdown, there will be another section such as Profile and Logout. Students will be redirected to the Guest homepage if they clicked on the Logout option.

VIEW YOUR COURSES

You can only modify next semester course registration.

Select Semester:

Semester 2024/2025-2

TRANSACTION ID	SEMESTER	COURSE	COURSE NAME	STATUS	ACTIONS
36	2024/2025-2	SECP3223	Data Analytics Programming	Accepted	<button>MODIFY</button> <button>CANCEL</button>

adriana.great-site.net/studentprofile.php CRS developed by Adriana @ 2024

8. If a student clicked on the Profile section, it would display their information. They will also be able to change their profile information.

WELCOME, MARIA CHEW!

Full Name

Maria Chew

Email

maria.chew@gmail.com

Contact Number

346921657

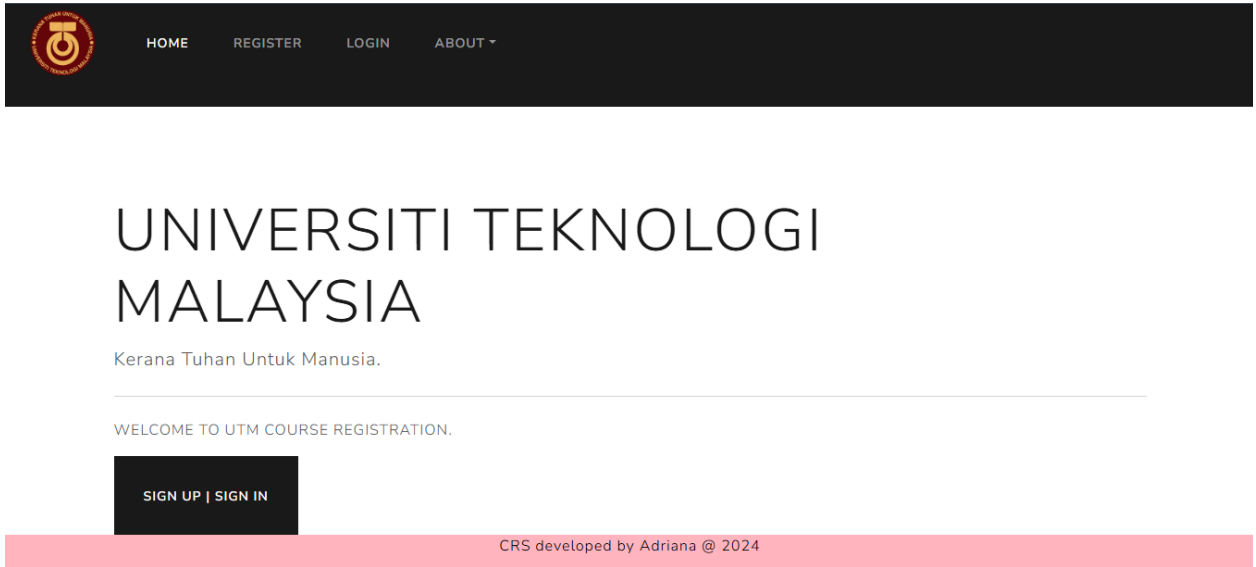
State

Terengganu

CRS developed by Adriana @ 2024

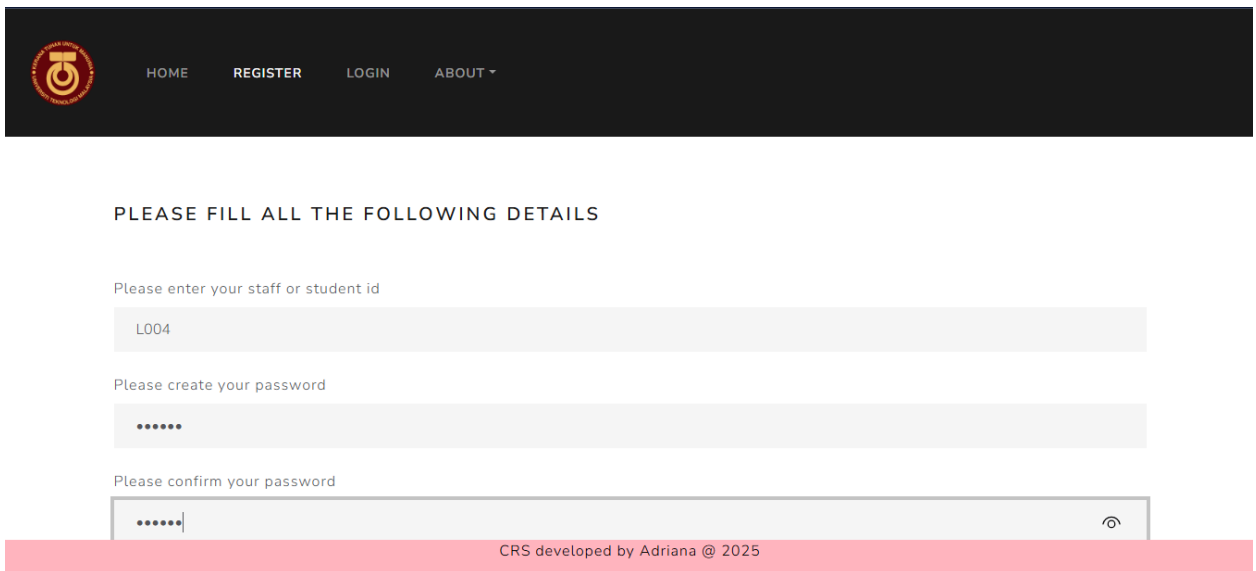
For Lecturer ;

9. At the guest homepage, select the Register section for new users.



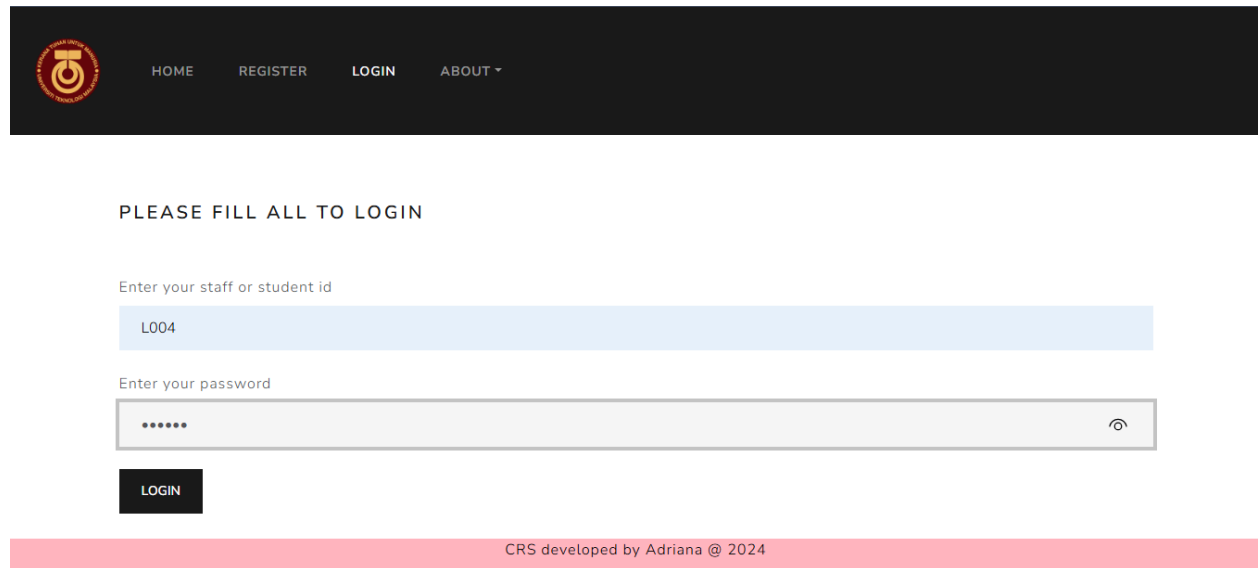
The screenshot shows the homepage of the UTM Course Registration System (CRS). At the top is a dark navigation bar with the UTM logo on the left and links for HOME, REGISTER, LOGIN, and ABOUT on the right. Below the navigation bar, the text "UNIVERSITI TEKNOLOGI MALAYSIA" is displayed in large, bold, black letters. Underneath this, the tagline "Kerana Tuhan Untuk Manusia." is shown. A horizontal line separates this from the text "WELCOME TO UTM COURSE REGISTRATION." Below this text is a black button with the text "SIGN UP | SIGN IN" in white. At the bottom of the page, a pink footer bar contains the text "CRS developed by Adriana @ 2024".

10. Fill in the Registration Form as new users.



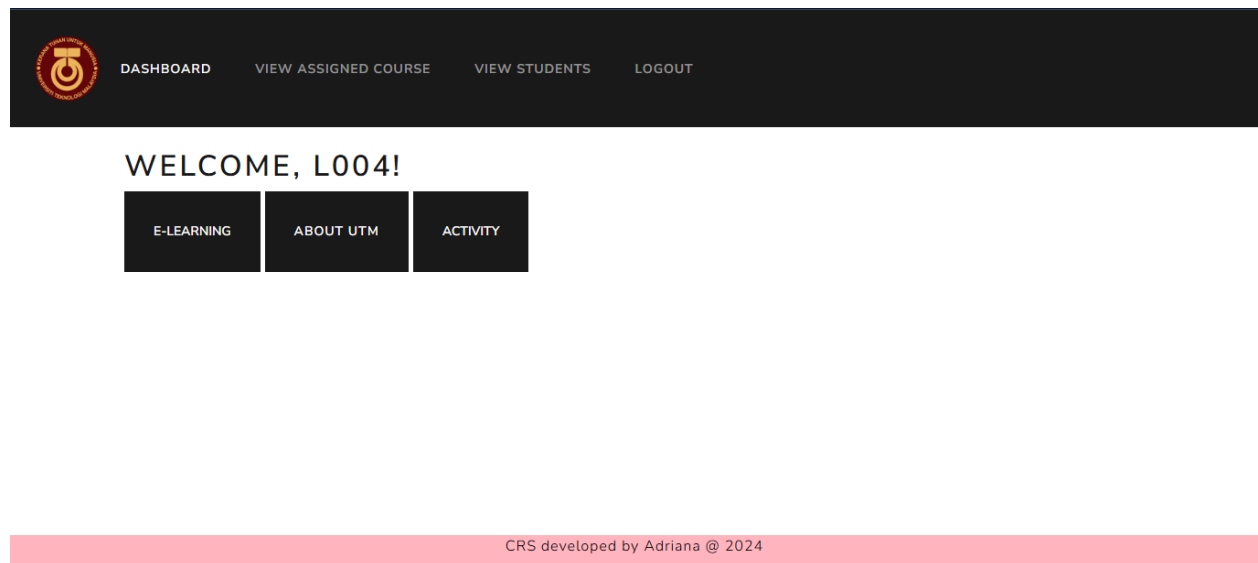
The screenshot shows the registration form for new users. At the top is a dark navigation bar with the UTM logo on the left and links for HOME, REGISTER, LOGIN, and ABOUT on the right. Below the navigation bar, the text "PLEASE FILL ALL THE FOLLOWING DETAILS" is displayed. The form consists of three input fields: 1. "Please enter your staff or student id" with the text "L004" entered. 2. "Please create your password" with a masked password ".....". 3. "Please confirm your password" with a masked password "....." and a toggle icon on the right. At the bottom of the page, a pink footer bar contains the text "CRS developed by Adriana @ 2025".

11. The page will redirect to the Login page once registration is successful.



The screenshot shows a login page with a dark header. On the left is the UTM logo. To its right are navigation links: HOME, REGISTER, LOGIN, and ABOUT with a dropdown arrow. The main content area is white and contains the text "PLEASE FILL ALL TO LOGIN". Below this are two input fields: "Enter your staff or student id" with the value "L004" and "Enter your password" with masked characters ".....". A "LOGIN" button is positioned below the password field. At the bottom, a pink footer bar contains the text "CRS developed by Adriana @ 2024".

12. This is the homepage of every lecturer.



The screenshot shows a lecturer dashboard with a dark header. On the left is the UTM logo. To its right are navigation links: DASHBOARD, VIEW ASSIGNED COURSE, VIEW STUDENTS, and LOGOUT. The main content area is white and displays "WELCOME, L004!". Below this are three buttons: E-LEARNING, ABOUT UTM, and ACTIVITY. At the bottom, a pink footer bar contains the text "CRS developed by Adriana @ 2024".

13. Lecturers can see their assigned course and its details once they click on the View Assigned Course page.

DASHBOARDVIEW ASSIGNED COURSEVIEW STUDENTSLOGOUT

ASSIGNED COURSES

SEMESTER: 2024/2025-2

COURSE CODE	COURSE NAME	CREDIT HOURS	MAX STUDENTS
SECD3761	Technopreneurship Seminar	1	5
SECP2753	Data Mining	3	5
SECR2043	Operating Systems	3	5
ULRS1022	Philosophy & Current Issues	3	5

CRS developed by Adriana @ 2024

14. If lecturers click on the View Students page, they can see students that registered under lecturer's courses. At the most left is the logout section, that will redirect lecturers to guest homepage.

DASHBOARDVIEW ASSIGNED COURSEVIEW STUDENTSLOGOUT

REGISTERED STUDENTS

Select Semester:

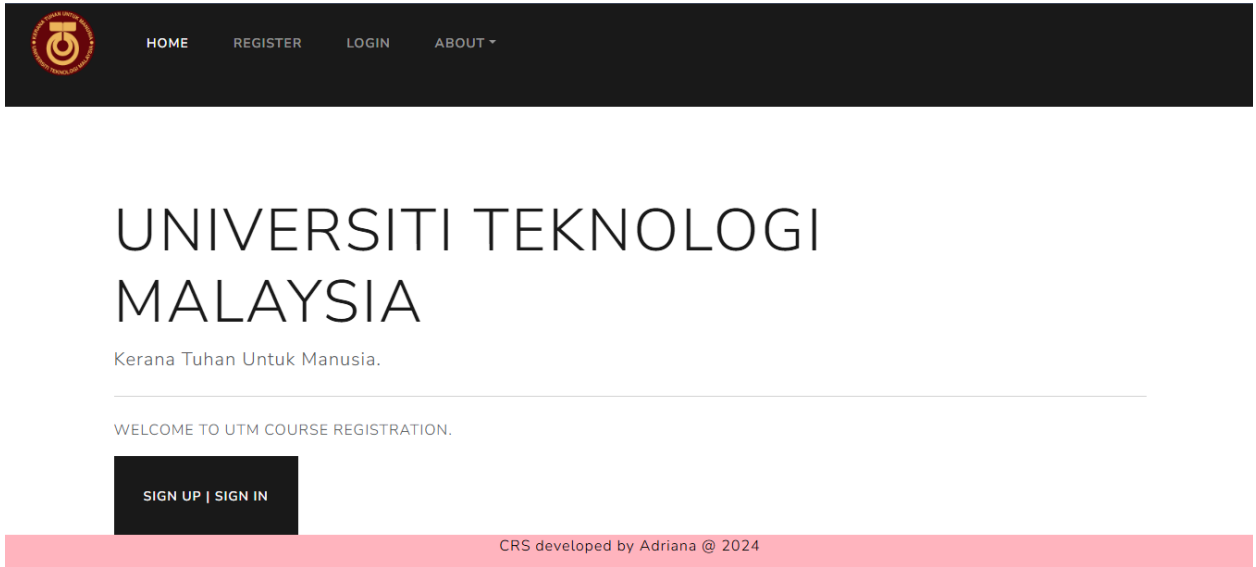
2024/2025-2

Course: Technopreneurship Seminar (SECD3761) - 1 students	^
Course: Data Mining (SECP2753) - 1 students	^
Course: Operating Systems (SECR2043) - 1 students	^
Course: Philosophy & Current Issues (ULRS1022) - 1 students	^

CRS developed by Adriana @ 2024

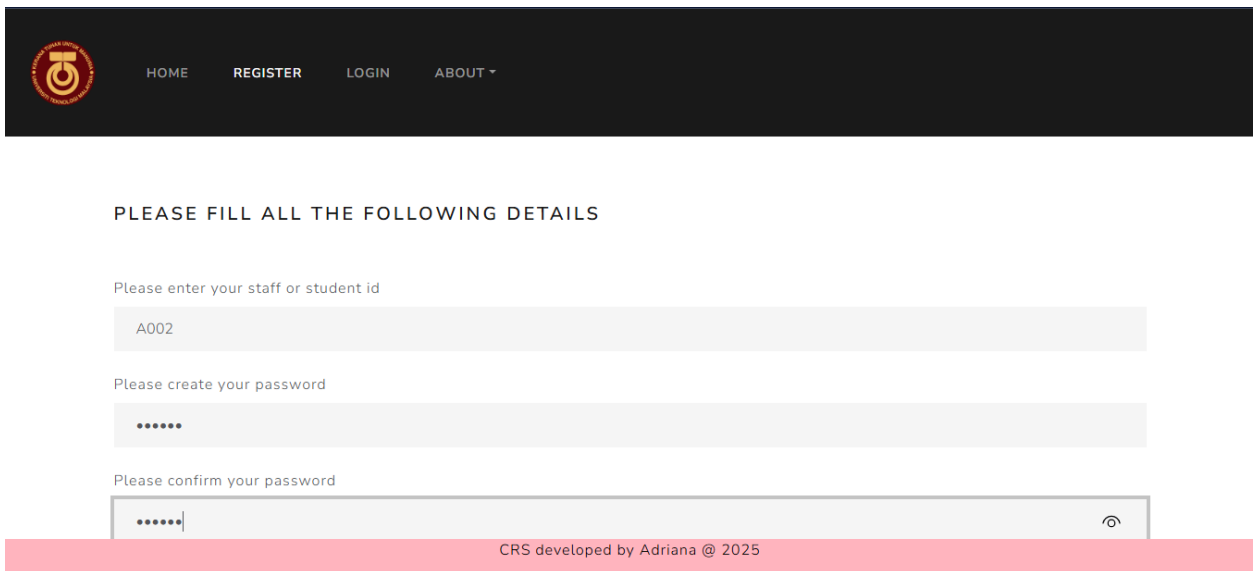
For Admin ;

15. At the guest homepage, select the Register section for new users.



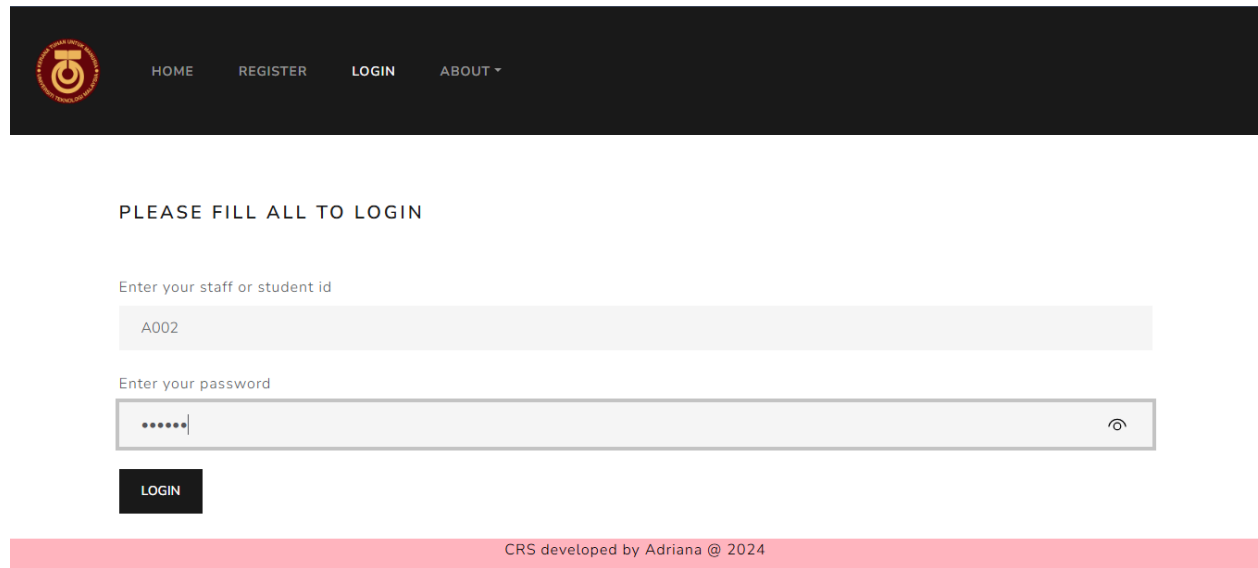
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16. Fill in the Registration Form as new users.



The screenshot shows the registration form of the UTM Course Registration System (CRS). At the top is a dark navigation bar with the UTM logo on the left and links for HOME, REGISTER, LOGIN, and ABOUT on the right. Below the navigation bar, the text "PLEASE FILL ALL THE FOLLOWING DETAILS" is displayed in bold, black letters. The form consists of three input fields: 1. "Please enter your staff or student id" with the text "A002" entered. 2. "Please create your password" with a masked password ".....". 3. "Please confirm your password" with a masked password "....." and a toggle icon (an eye) on the right. At the bottom of the page, a pink footer bar contains the text "CRS developed by Adriana @ 2025".

17. The page will redirect to the Login page once registration is successful.



PLEASE FILL ALL TO LOGIN

Enter your staff or student id

A002

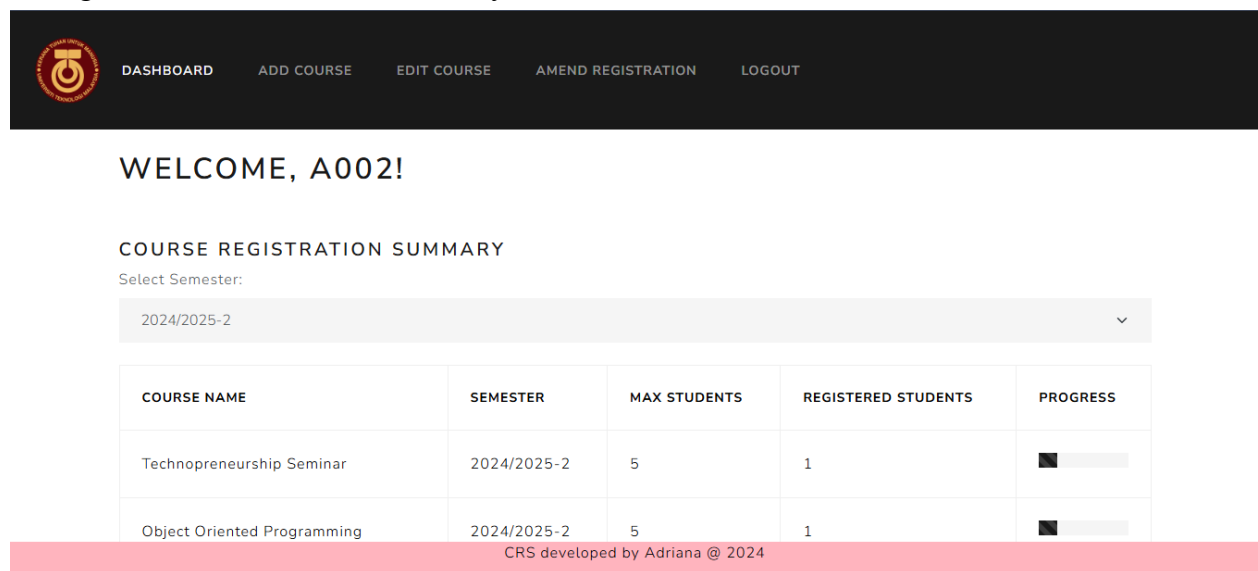
Enter your password

.....

LOGIN

CRS developed by Adriana @ 2024

18. The homepage of every admin will start with Dashboard. Here, it shows the current registration of students under every course.



WELCOME, A002!

COURSE REGISTRATION SUMMARY

Select Semester:

2024/2025-2

COURSE NAME	SEMESTER	MAX STUDENTS	REGISTERED STUDENTS	PROGRESS
Technopreneurship Seminar	2024/2025-2	5	1	
Object Oriented Programming	2024/2025-2	5	1	

CRS developed by Adriana @ 2024

19. Admin may add or edit courses. They may also Amend or Cancel Registration as shown below. The admin will be redirected to the guest homepage once they click on the Logout section.

ADD COURSES

Course Code

Course Name

Credit Hours

Lecturer ID

CRS developed by Adriana @ 2024

EDIT COURSE

Admin may edit, update, or delete courses here.

Select Semester:

2024/2025-2

▼

SECD3761 - Technopreneurship Seminar

▼

SECJ2154 - Object Oriented Programming

▼

SECP2753 - Data Mining

▼

SECP3223 - Data Analytics Programming

▼

CRS developed by Adriana @ 2024

AMEND OR CANCEL REGISTRATION

Admin may amend or delete course registration here.

Select Semester:

2024/2025-2

▼

STUDENT ID	COURSE NAME	AMEND TO	CANCEL
S001	Technopreneurship Seminar	Technopreneurship Seminar ▼ UPDATE	CANCEL

CRS developed by Adriana @ 2024

7.0 LOCALHOST SETUP

1. Download and install XAMPP Control Panel from <https://www.apachefriends.org/index.html>
2. Start Apache and MySQL.

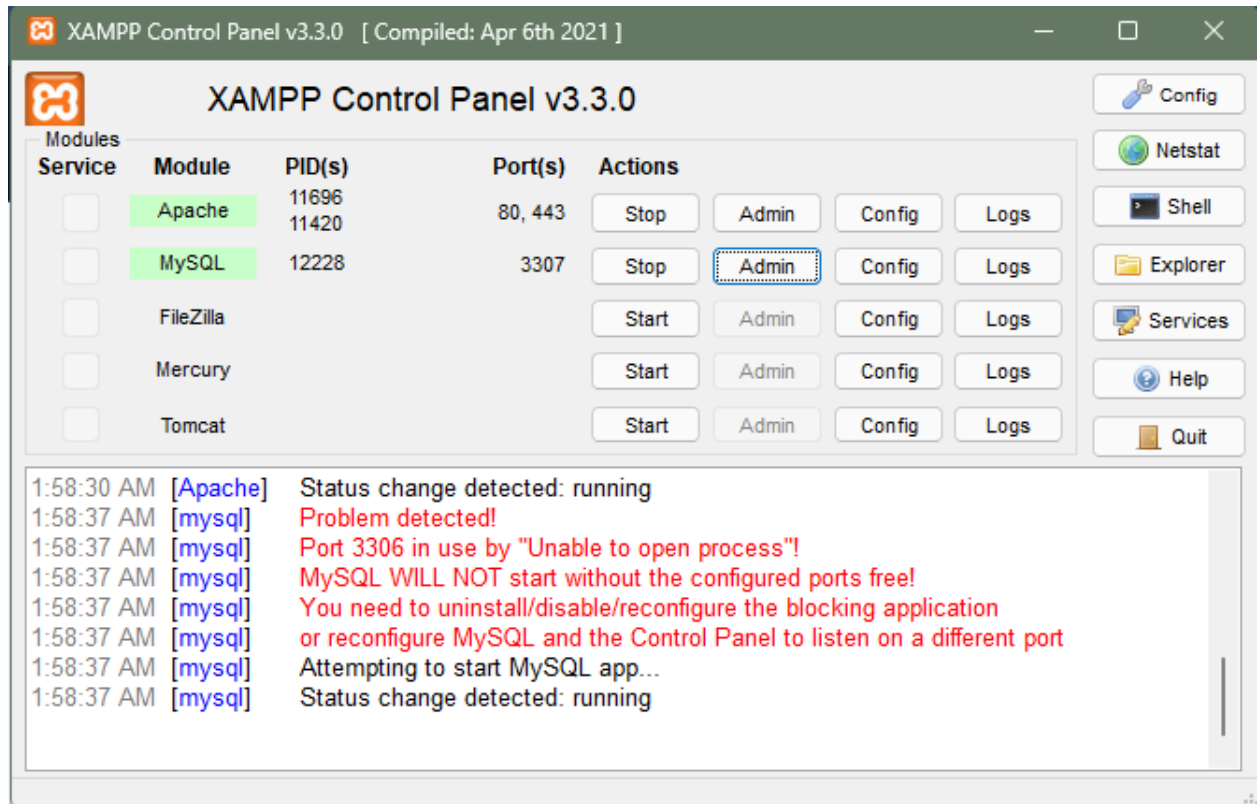


Figure 7.0.1 XAMPP Control Panel

3. Click the “Admin” button in MySQL module’s “Actions” section to view phpMyAdmin.
4. Open the crs.zip file and extract them into OS(C:)>xampp> ‘htdocs’ file path.

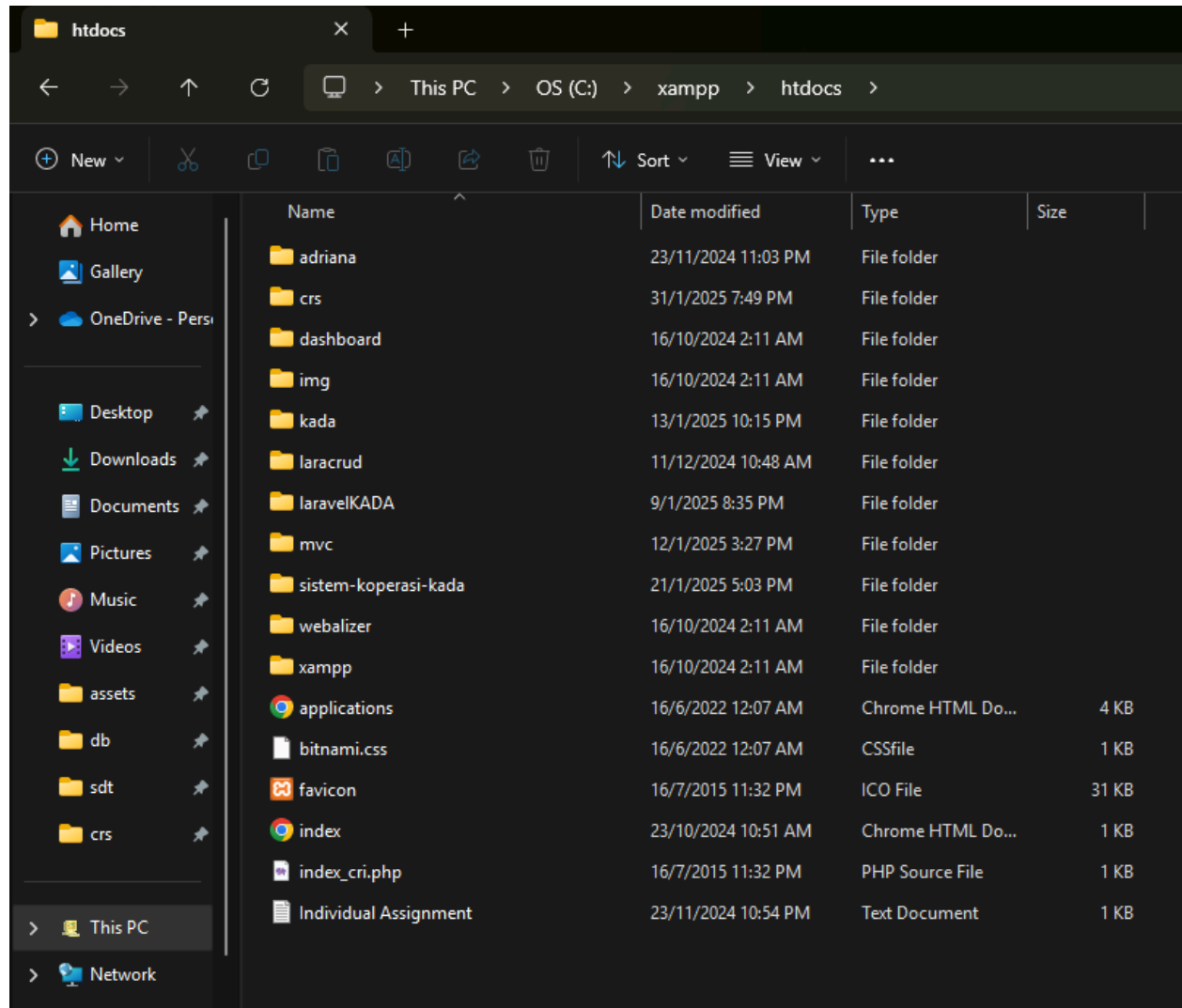


Figure 7.0.2 'htdocs' file path

5. Create a new database called 'db_crs'.

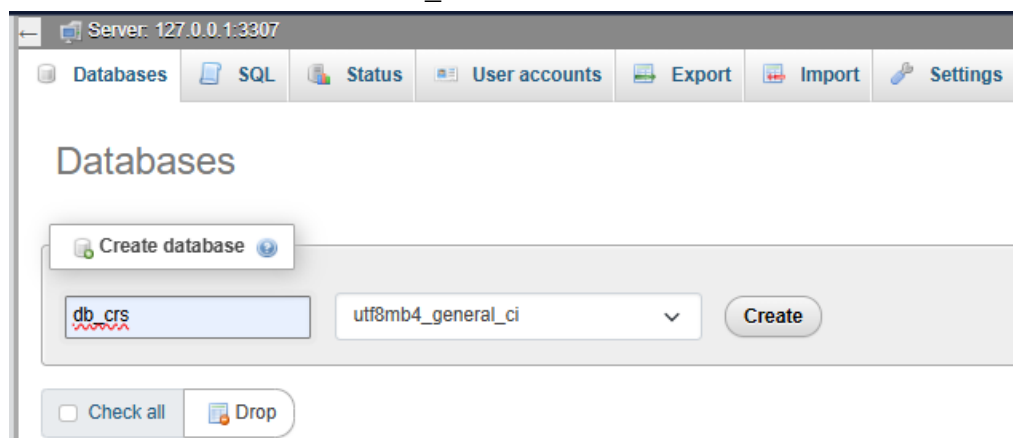


Figure 7.0.3 Create Database

6. Import 'db_crs.sql' from the folder 'crs' to database 'db_crs'.

Importing into the database "db_crs"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **.[format].[compression]**. Example: **.sql.zip**

Browse your computer: (Max: 40MiB)

Choose File

db_crs.sql

You may also drag and drop a file on any page.

Character set of the file:

utf-8

Figure 7.0.4 Import Database

8.0 HOSTING URL

Course Registration System OR <http://adriana.great-site.net/index.php>

9.0 USER CREDENTIALS

User ID	Password
A001	342516
A002	342516
L003	654321
L004	654321
S001	123456
S002	123456
S003	123456
S004	123456
S005	123456
S006	123456
S007	123456